NATURAL LANGUAGE PROCESSING

Question Bank with Options and Correct Answers

Units 1, 2 and 3

UNIT – 1

Choose the correct alternative

1. What is the main focus of morphological analysis in NLP?

a) Sentence structure
b) Document classification
c) Word structure
d) Syntax parsing

Answer: c) Word structure

2. Which of the following is NOT a type of morpheme?

a) Free morpheme
b) Bound morpheme
c) Infinite morpheme
d) Derivational morpheme

Answer: c) Infinite morpheme

3. Which technique is used to find the root form of a word?

a) Tokenization
b) Stemming
c) POS tagging
d) Sentence parsing

Answer: b) Stemming

4. What is a major drawback of rule-based morphological models?

a) High accuracy
b) Scalability issues
c) Requires deep learning
d) Fast processing

Answer: b) Scalability issues

5. What is the main challenge in morphological analysis?

a) Low memory usage
b) Ambiguity
c) Simple rules
d) No computation

Answer: b) Ambiguity

Fill in the blanks (Unit–1)

1. The smallest unit of meaning is called a **morpheme**.

2. A morpheme that can stand alone is a **free** morpheme.

3. Reducing words to base form is called **lemmatization**.

4. An affix added at the beginning is called a **prefix**.

5. **Stemming** does not always produce valid dictionary words.

UNIT – 2

Choose the correct alternative

1. What is the main goal of syntax analysis in NLP?

a) Word segmentation
b) Identifying sentence structure
c) Speech recognition
d) Tokenization

Answer: b) Identifying sentence structure

2. Which parsing technique builds parse trees bottom-up?

a) Top-down parsing
b) Bottom-up parsing
c) Rule-based parsing
d) Random parsing

Answer: b) Bottom-up parsing

3. Dependency parsing represents:

a) Constituency structure
b) Relationships between words
c) Word embeddings
d) Tokenization

Answer: b) Relationships between words

4. Which corpus is used for syntactic parsing?

a) WordNet
b) ImageNet
c) Penn Treebank
d) BERT

Answer: c) Penn Treebank

5. Which algorithm is used for probabilistic parsing?

a) Naive Bayes
b) CKY Algorithm
c) K-means
d) A*

Answer: b) CKY Algorithm

Fill in the blanks (Unit–2)

1. Parsing analyzes the **syntactic** structure of a sentence.

2. Root to leaf parsing is called **top-down** parsing.

3. Leaf to root parsing is called **bottom-up** parsing.

4. Dependency parsing forms a **directed** graph.

5. **CKY** algorithm parses PCFGs.

UNIT – 3

Choose the correct alternative

1. What is the main objective of semantic parsing?

a) Speech recognition
b) Syntactic analysis
c) Convert text to structured meaning
d) Tokenization

Answer: c) Convert text to structured meaning

2. A major challenge in semantic parsing is:

a) Tokenization
b) Ambiguity in meaning
c) POS tagging
d) Regex

Answer: b) Ambiguity in meaning

3. Which method is used for word sense disambiguation?

a) Bag of Words
b) Knowledge-based methods
c) Image recognition
d) Parsing

Answer: b) Knowledge-based methods

4. Which model is widely used in modern semantic parsing?

a) Decision Trees
b) CNNs
c) Transformers
d) Naive Bayes

Answer: c) Transformers

5. Rule-based semantic parsing uses:

a) Neural networks
b) Statistical models
c) Linguistic rules
d) Attention models

Answer: c) Linguistic rules

Fill in the blanks (Unit–3)

1. Structural ambiguity occurs at the **syntactic** level.

2. Ambiguity arises due to **polysemy** and homonymy.

3. **Semantic parsing** converts language to machine-readable form.

4. **Hybrid** approaches combine rules and ML.

5. **Transformer-based** models capture context.